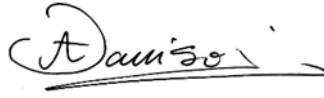


Approved by Chair:



Jan 2, 2025

Signature

COURSE SECTION INFORMATION

**Mathematical Concepts for Deep Learning
Applied A.I. Solutions Development**

Note: All academic inquiries will be replied to within three business days.

COURSE DESCRIPTION:

This course will provide an introduction to the mathematical fundamentals of Deep Learning algorithms. The main focus will be on the exploration and implementation of Deep Learning algorithms for text and audio analysis. Students will gain practical experience with popular mathematical tools, techniques, and libraries in a variety of use cases.

COURSE OUTCOMES:

Upon successful completion of this course the students will have reliably demonstrated the ability to:

1. Interpret mathematics of Deep Learning algorithms.
2. Apply concepts such as Vector Calculus, and Matrix Algebra utilising widely-used mathematical libraries.
3. Apply Deep Learning algorithms (NLP, RNN, Transformers...).

LIST OF TEXTBOOKS AND OTHER TEACHING AIDS:

Required:

Given the fast pace of the field, the course is based on multiple resources. The key resources are listed below. None is a required reading, although they are recommended as an overview. All required content will be delivered as part of the lectures.

Recommended Resources:

- **Deep Learning with Python**, François Chollet (2nd Edition, October 2021), [Manning](#), [Github repo](#).

- **Natural Language Processing with Transformers**, Lewis Tunstall, Leandro von Werra, Thomas Wolf (revised edition, June 2022), [O'reilly](#), [Github repo](#).
- **Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow**, Aurélien Géron (October 2022), [O'Reily](#), [Github repo](#).
- **Deep Learning with PyTorch**, Eli Stevens, Luca Antiga, and Thomas Viehmann (July 2020), [Manning](#), [Github repo](#)
- **Machine Learning with PyTorch and Scikit-Learn**, Sebastian Raschka et al. (February 2022), [Pckt](#), [Github repo](#)

Important Note on the Use of Generative AI:

Students must review the "Generative AI Usage Guidelines" document, available on D2L, or consult with the instructor for details on how generative AI tools may be used in this course.

Generally, use of AI is allowed and encouraged in most of the courses of this program, this include projects and assignments. However use of AI are not allowed in exams and quizzes unless specified otherwise by the instructor. Students must consult their instructor when unsure.

Misuse of AI in assessments where it is not permitted or failure to adequately disclose its use will be treated as a violation of academic integrity. According to college policy, consequences may include failing the assignment or the course or more severe disciplinary actions. **Students must also download the AI Usage Declaration form from D2L, complete it, and submit it with their assignments where AI use is permitted.** Adherence to these guidelines is mandatory to maintain academic integrity.

Detailed Evaluation System

Assessment Tool:	Description:	Outcome(s) assessed:	EES assessed:	Date / Week:	% of Final Grade:
Participation	Active participation in class, in-person	1, 2, 3, 4, 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	1	10
Mid-term Exam	This is an in-class exam that will evaluate the knowledge of the student. The format is a multiple-choice questionnaire.	1, 2, 3, 4, 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	2	20
Final Exam	This is an in-class exam that will evaluate the knowledge of the student. The format is a multiple-choice questionnaire.	1, 2, 3, 4, 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	4	30
Project and Presentation	Assignment in pairs	1, 2, 3, 4, 5	1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11	4	40
TOTAL:					100%

GRADING SYSTEM the passing grade for this course is: D (50%)

A+	90-100	4.0	B+	77-79	3.3	C+	67-69	2.3	D+	57-59	1.3	Below 50	F	0.0
A	86-89	4.0	B	73-76	3.0	C	63-66	2.0	D	50-56	1.0			
A-	80-85	3.7	B-	70-72	2.7	C-	60-62	1.7						

Excerpt from the College Policy on Academic Dishonesty:

The *minimal* consequence for submitting a plagiarized, purchased, contracted, or in any manner inappropriately negotiated or falsified assignment, test, essay, project, or any evaluated material will be a grade of zero on that material.

To view George Brown College policies please go to www.georgebrown.ca/policies

Learning Schedule / Topical Outline (subject to change with notification)
--

TOPICAL OUTLINE:

Week	Topic Task	In-person / On-line	Outcome(s)	Content / Activities	Resources
1	1	Tuesday In-Person	1, 2, 3	- A brief definition of deep-learning - Demonstration of deep learning capabilities in NLP tasks - Setting up a development environment for deep learning - Tensor arithmetic - Linear regression model - Gradient descent	Resource material available on D2L.
1	2	Thursday Online	1, 2, 3	- Activation functions: sigmoid, ReLU - The perceptron - Feedforward neural network - The loss function - Backpropagation - Fitting and overfitting - Learning rate - Training a binary classifier	Resource material available on D2L
					Week 1 Tasks: Lab Exercise 1
2	3	Tuesday In-Person	1, 2, 3, 4	- Training, validation, and test sets - Data normalization - Loss function: L1, L2 regularization - Optimizers: Momentum and Adaptive momentum (Adam) - Confusion metric, precision, accuracy and ROC curve - Hyperparameters-tuning, grid search - Multi-class classifier	Resource material available on D2L
2	4	Thursday Online	1, 2, 3, 4	- Vector spaces and dimensionality reduction - Word Embeddings (word2vec, GLoVe) - Evaluation metrics for NLP	Resource material available on D2L
					Week 2 Tasks: Home Assignment 2, Quiz
3	5	Tuesday In-Person	1, 2, 3, 4, 5	- Recurrent Neural Networks (RNN) - vanilla RNN - Backpropagation-through-time (BPTT) - Text Tokenization	Resource material available on D2L
3	6	Thursday Online	1, 2, 3, 4, 5	- The LSTM cell - Encoder-Decoder LSTM and sequence-to-sequence problems - The GRU cell - Dropout and Regularization	Resource material available on D2L
					Week 3 Tasks: Home Assignment 3 and 4
4	7	Tuesday In-Person	1, 2, 3, 4, 5,	- Introduction to Transformers - The query-key-value decomposition - Self-attention, scaled dot-product attention - Multi-head attention - Positional embeddings - Skip connections	Resource material available on D2L

				- Layer normalization	
4	8	Thursday Online	1, 2, 3, 4, 5	- The Encoder, Decoder and Encoder-Decoder architectures (BERT, GPT) - Transformers for NLP tasks (sentiment analysis, named entity recognition, question answering) - Fine-tuning transformers (transfer learning) - Understanding how to train a transformer model from scratch	Resource material available on D2L
					Week 4 Tasks: Final Exam, Project Due
				Please note: this schedule may change as resources and circumstances require. For information on withdrawing from this course without academic penalty, please refer to the College Academic Calendar: http://www.georgebrown.ca/Admin/Registr/PSCal.aspx	